Joshua Ahlers

WORK EXPERIENCE

Northwestern Mutual, Milwaukee, WI
03/19/2024 - Present
Data Engineer

- Productionized machine learning models by refactoring data science notebooks and scripts into reusable, testable Python modules aligned with internal MLOps standards.
- Migrated six marketing models from Domino to Databricks, building training and prediction pipelines with Python, PySpark, MLflow, XGBoost, and Databricks workflows.
- Enhanced internal MLOps stack templates with reusable support for XGBoost, CatBoost, Spark, scikit-learn, classification, regression, and Databricks Feature Store.
- Re-architected the MLOps stack into a plugin-based framework, enabling teams to extend algorithms, data frameworks, feature stores, and problem types without modifying core code.
- Built CLI and Databricks-hosted UI tooling for generating MLOps project scaffolds through guided prompts, reusable templates, and automated project setup.
- Developed GenAI tooling that converts Jupyter notebook logic into reusable Python functions, generates pytest test suites, and supports notebook-to-production refactoring.
- Created proof-of-concepts for real-time data streaming, RAG stack plugins, MCP Gateway authentication, and Databricks resource cleanup automation.
- Mentored data scientists on Databricks, MLflow, deployment workflows, and MLOps best practices through hands-on support and recurring working sessions.

Northwestern Mutual, Franklin, WI
11/14/2023 - 12/29/2023
Application Engineer

- Designed, developed, and implemented APIs that align with product owners' goals and standards.
- Developed several toggles and email timers for domain owners to help efficiently notify their team that their API needed updates.
- Worked across teams using Tyk, AWS, and Kubernetes.
- Oversaw the entire API lifecycle, from design and development to deployment and maintenance.
- Participated in code reviews and provided feedback on pipeline issues.

Northwestern Mutual, Milwaukee, WI
05/02/2023 - 10/06/2023
iOS Mobile Engineer

- Developed and maintained unit tests for iOS applications.
- Successfully refactored the architecture from Model-View-ViewModel (MVVM) to The Composable Architecture (TCA), improving code maintainability and scalability.
- Seamlessly integrated GraphQL into the iOS application, enhancing data retrieval and management capabilities.

- Participated in code reviews and provided constructive feedback to enhance code quality and team collaboration.
- Enhanced the user interface of iOS applications by integrating and refining UI elements, contributing to a more visually appealing and user-friendly design.

EDUCATION

UWM, Milwaukee, WI
09/01/2017 - 08/31/2021
GPA: 3.73

- BA, Double Major in Economics and Philosophy.
- Awards: Magna Cum Laude, Commencement Honors.
- Relevant coursework: Economic Applications of Game Theory, Symbolic Logic, Health Economics, Economic Employment & Labor Relations, Linguistic Analysis.
- Relevant research: Research in Economic Game Theory: Trust Games Between Companies, Research in Symbolic Logic: Symbolic Logic Proofs and Validity.

Boston University, Online
08/28/2026 – Current

- Graduate Degree in Computer Science and Artificial Intelligence.
- Relevant coursework: Foundations of Programming and Systems, Mathematical Foundations of Machine Learning.

SELECTED PROJECTS

- **MLOps Stack Plugin Architecture**: Designed a plugin-based Python framework for internal MLOps project generation, enabling support for multiple algorithms, data
- **Notebook-to-Function GenAI Tool**: Built an LLM-powered tool that scans Jupyter notebooks, extracts reusable logic, generates Python modules, and creates pytest suites to accelerate notebook-to-production workflows.
- **MCP Gateway POC**: Developed a secure gateway architecture for MCP resource servers using Azure Entra ID token validation, centralized request routing, and zero-trust downstream authentication.

SKILLS

- **Languages**: Python, Java, JavaScript, TypeScript, Swift, SQL, HTML/CSS
- **Data Engineering & M**L: PySpark, pandas, NumPy, scikit-learn, XGBoost, CatBoost, MLflow, Databricks, Databricks Feature Store, Ray Train, Ray Tune
  MLOps & Platforms: Databricks Workflows, Databricks Asset Bundles, model deployment, model serving, plugin architectures, Cookiecutter, Click, pytest, Jupyter notebooks
- **Backend & APIs**: FastAPI, Flask, Spring Framework, REST APIs, GraphQL, node.js
- **Frontend**: React, Next.js, SwiftUI, Bootstrap
- **Databases & Streaming**: MySQL, PostgreSQL, Kafka, Redis
- **DevOps & Cloud**: Docker, Docker Compose, Kubernetes, GitLab CI/CD, AWS, Terraform,Liquibase
- **Observability & Testing**: Jaeger, Locust, pytest, unit testing, load testing

- **GenAI & Agentic Tools**: LLM-powered code generation, code review agents, RAG stack architecture, MCP Gateway POCs